

HG-KR Series (Low Inertia, Small Capacity) Specifications

Rotary servo motor model			HG-KR	053(B)	13(B)	23(B)	43(B)	73(B)
Compatible servo amplifier model			MR-J4-	Refer to "Combinations of Rotary Servo Motor and Servo Amplifier (200 V Class)" on p. 2-3 in this catalog.				
			MR-J4W_-					
Power supply capacity ^{*1}			[kVA]	0.3	0.3	0.5	0.9	1.3
Continuous running duty	Rated output	[W]	50	100	200	400	750	
	Rated torque ^(Note 3)	[N•m]	0.16	0.32	0.64	1.3	2.4	
Maximum torque			[N•m]	0.56	1.1	2.2	4.5	8.4
Rated speed			[r/min]	3000				
Maximum speed			[r/min]	6000				
Permissible instantaneous speed			[r/min]	6900				
Power rate at continuous rated torque	Standard	[kW/s]	5.63	13.0	18.3	43.7	45.2	
	With electromagnetic brake	[kW/s]	5.37	12.1	16.7	41.3	41.6	
Rated current			[A]	0.9	0.8	1.3	2.6	4.8
Maximum current			[A]	3.2	2.5	4.6	9.1	17
Regenerative braking frequency ^{*2}	MR-J4-	[times/min]	(Note 4)	(Note 4)	453	268	157	
	MR-J4W_-	[times/min]	2500	1350	451	268	393	
Moment of inertia J	Standard	[× 10 ⁻⁴ kg•m ²]	0.0450	0.0777	0.221	0.371	1.26	
	With electromagnetic brake	[× 10 ⁻⁴ kg•m ²]	0.0472	0.0837	0.243	0.393	1.37	
Recommended load to motor inertia ratio ^(Note 1)			17 times or less		26 times or less	25 times or less	17 times or less	
Speed/position detector			Absolute/incremental 22-bit encoder (resolution: 4194304 pulses/rev)					
Oil seal			None	None (Servo motors with oil seal are available. (HG-KR_J))				
Insulation class			130 (B)					
Structure			Totally enclosed, natural cooling (IP rating: IP65) ^(Note 2)					
Environment ^{*3}	Ambient temperature	0 °C to 40 °C (non-freezing), storage: -15 °C to 70 °C (non-freezing)						
	Ambient humidity	80 %RH maximum (non-condensing), storage: 90 %RH maximum (non-condensing)						
	Ambience	Indoors (no direct sunlight); no corrosive gas, inflammable gas, oil mist or dust						
	Altitude	1000 m or less above sea level						
	Vibration resistance ^{*4}	X: 49 m/s ² Y: 49 m/s ²						
Vibration rank			V10 ^{*6}					
Compliance to standards			Refer to "Conformity with global standards and regulations" on p. 30 in this catalog.					
Permissible load for the shaft ^{*5}	L	[mm]	25	25	30	30	40	
	Radial	[N]	88	88	245	245	392	
	Thrust	[N]	59	59	98	98	147	
Mass	Standard	[kg]	0.34	0.54	0.91	1.4	2.8	
	With electromagnetic brake	[kg]	0.54	0.74	1.3	1.8	3.8	

Notes: 1. Contact your local sales office if the load to motor inertia ratio exceeds the value in the table.

2. The shaft-through portion is excluded. For geared servo motor, IP rating of the reducer portion is equivalent to IP44. Refer to the asterisk 7 of "Annotations for Rotary Servo Motor Specifications" on p. 2-27 in this catalog for the shaft-through portion.

3. When unbalanced torque is generated, such as in a vertical lift machine, it is recommended that the unbalanced torque of the machine be kept under 70% of the servo motor rated torque.

4. When the servo motor decelerates to a stop from the rated speed, the regenerative frequency will not be limited if the effective torque is within the rated torque range.

When the servo motor decelerates to a stop from the maximum speed, the regenerative frequency will not be limited if the following requirements are met.

• HG-KR053(B): The load to motor inertia ratio is 8 times or less, and the effective torque is within the rated torque range.

• HG-KR13(B): The load to motor inertia ratio is 4 times or less, and the effective torque is within the rated torque range.

Refer to "Annotations for Rotary Servo Motor Specifications" on p. 2-27 in this catalog for the asterisks 1 to 6.

HG-KR Series Electromagnetic Brake Specifications (Note 1)

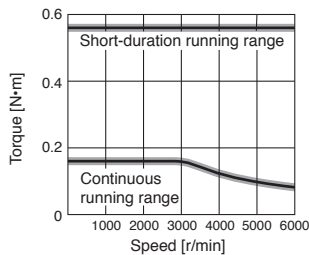
Model	HG-KR	053B	13B	23B	43B	73B
Type	Spring actuated type safety brake					
Rated voltage	24 V DC _{-10%}					
Power consumption [W] at 20 °C		6.3	6.3	7.9	7.9	10
Electromagnetic brake static friction torque [N·m]		0.32	0.32	1.3	1.3	2.4
Permissible braking work	Per braking [J]	5.6	5.6	22	22	64
	Per hour [J]	56	56	220	220	640
Electromagnetic brake life (Note 2)	Number of brakings [Times]	20000	20000	20000	20000	20000
	Work per braking [J]	5.6	5.6	22	22	64

Notes: 1. The electromagnetic brake is for holding. It should not be used for deceleration applications.

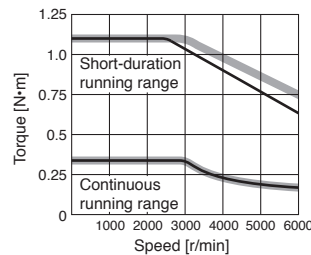
2. Brake gap is not adjustable. Electromagnetic brake life is defined as the time period until the readjustment is needed.

HG-KR Series Torque Characteristics (Note 3)

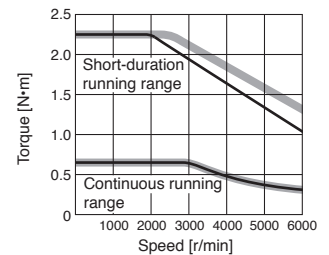
HG-KR053(B) (Note 1, 2)



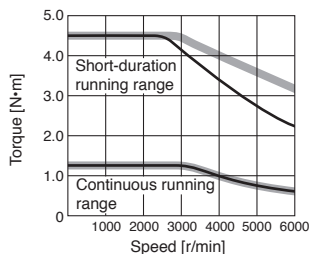
HG-KR13(B) (Note 1, 2)



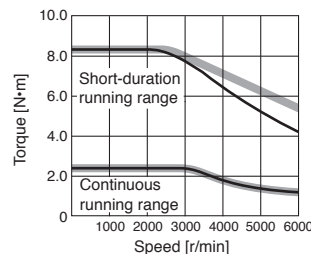
HG-KR23(B) (Note 1, 2)



HG-KR43(B) (Note 1, 2)



HG-KR73(B) (Note 1, 2)

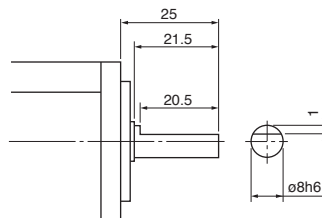


- Notes: 1. — : For 3-phase 200 V AC or 1-phase 230 V AC.
 2. — : For 1-phase 200 V AC.
 3. Torque drops when the power supply voltage is below the specified value.

HG-KR Series Special Shaft End Specifications

Motors with the following specifications are also available.

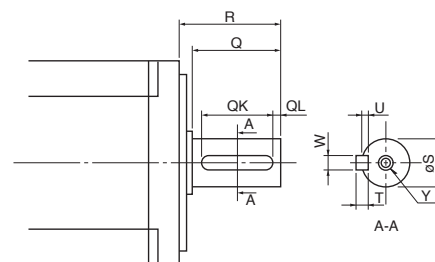
D-cut shaft (Note 1): 50 W and 100 W



[Unit: mm]

Key shaft (with key) (Note 1, 2): 200 W, 400 W, and 750 W

Model	Variable dimensions								
	T	S	R	Q	W	QK	QL	U	Y
HG-KR23(B)K, 43(B)K	5	14h6	30	26	5	20	3	3	M4 screw Depth: 15
HG-KR73(B)K	6	19h6	40	36	6	25	5	3.5	M5 screw Depth: 20



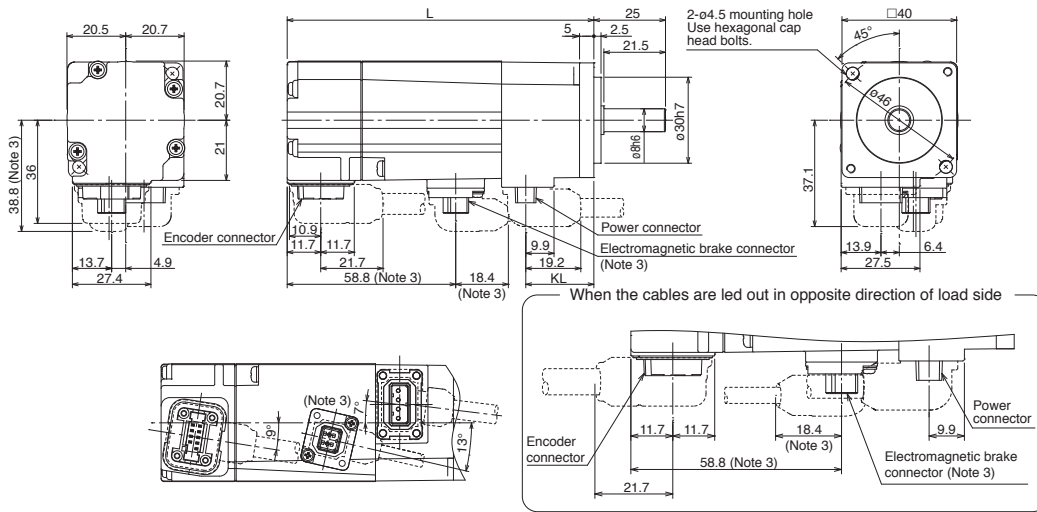
[Unit: mm]

Notes: 1. The servo motors with special shaft end are not suitable for frequent start/stop applications.

2. 2 round end key is attached.

HG-KR/HG-MR Series Dimensions (Note 1, 5, 6)

- HG-KR053(B), HG-KR13(B)
- HG-MR053(B), HG-MR13(B)



Power connector



Pin No.	Signal name
1	⊕ (PE)
2	U
3	V
4	W

Electromagnetic brake connector (Note 2)

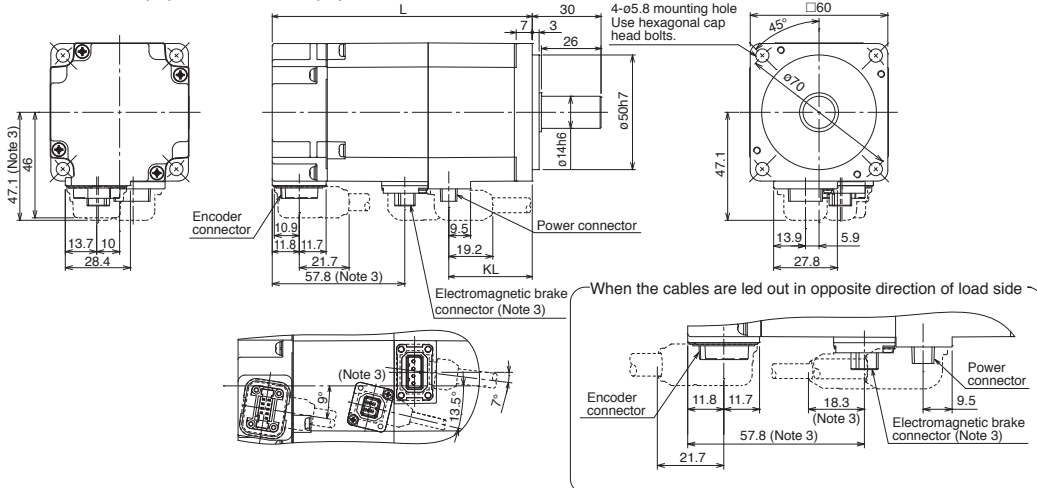


Pin No.	Signal name
1	B1
2	B2

Model	Variable dimensions (Note 4)	
	L	KL
HG-KR053(B) HG-MR053(B)	66.4 (107)	23.8
HG-KR13(B) HG-MR13(B)	82.4 (123)	39.8

[Unit: mm]

- HG-KR23(B), HG-KR43(B)
- HG-MR23(B), HG-MR43(B)



Power connector



Pin No.	Signal name
1	⊕ (PE)
2	U
3	V
4	W

Electromagnetic brake connector (Note 2)

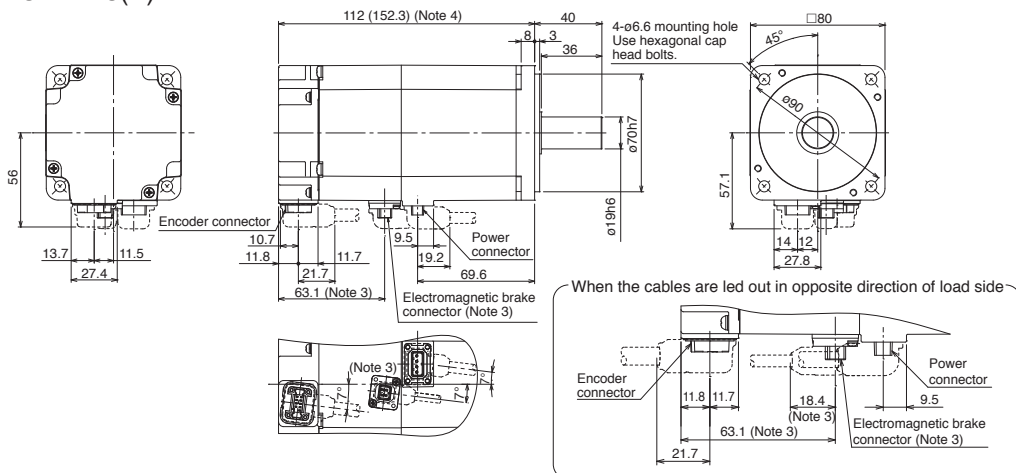


Pin No.	Signal name
1	B1
2	B2

Model	Variable dimensions (Note 4)	
	L	KL
HG-KR23(B) HG-MR23(B)	76.6 (113.4)	36.4
HG-KR43(B) HG-MR43(B)	98.3 (135.1)	58.1

[Unit: mm]

- HG-KR73(B)
- HG-MR73(B)



Power connector



Pin No.	Signal name
1	⊕ (PE)
2	U
3	V
4	W

Electromagnetic brake connector (Note 2)



Pin No.	Signal name
1	B1
2	B2

[Unit: mm]

Notes: 1. For dimensions without tolerance, general tolerance applies.

2. The electromagnetic brake terminals (B1, B2) do not have polarity.

3. Only for the models with electromagnetic brake.

4. Dimensions in brackets are for the models with electromagnetic brake.

5. Use a friction coupling to fasten a load.

6. Servo motors with oil seal (HG-KR_J and HG-MR_J) have different dimensions. Contact your local sales office for more details.